

Linear algebra – 8

Block matrices and determinants of special shape

12 сентября 2026 года

Theoretical minimum.

Def. For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with A invertible the *Schur complement* is $M/A = D - CA^{-1}B$, and $\det M = \det A \cdot \det(M/A)$; moreover $\text{rank } M = \text{rank } A + \text{rank}(M/A)$.

Theorem 1 (Sylvester). $\det(E_m + AB) = \det(E_n + BA)$ for $A \in M_{m,n}$, $B \in M_{n,m}$; consequently AB and BA have the same nonzero eigenvalues with the same multiplicities.

Theorem 2 (Cauchy–Binet). For $A \in M_{m,n}$, $B \in M_{n,m}$ with $m \leq n$, $\det(AB) = \sum_S \det A_{[m],S} \det B_{S,[m]}$, the sum over all m -element subsets $S \subseteq [n]$.

Fact 1. A circulant matrix with symbol $f(x) = b_0 + b_1x + \dots + b_{n-1}x^{n-1}$ has eigenvalues $f(\varepsilon^k)$, where $\varepsilon = e^{2\pi i/n}$; hence $\det C = \prod_{k=0}^{n-1} f(\varepsilon^k)$.

1. Prove the Schur determinant formula and the identity $\det(E + AB) = \det(E + BA)$; deduce that a block matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with commuting blocks C, D has determinant $\det(AD - BC)$, and give a counterexample when the blocks do not commute.
2. (IMC) Let M be an invertible $2n \times 2n$ matrix written in block form as $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $M^{-1} = \begin{pmatrix} E & F \\ G & H \end{pmatrix}$, all blocks of size $n \times n$. Show that $\det M \cdot \det H = \det A$.
3. (SEEMOUS 2018) Let $m, n, p, q \geq 1$ and let $A \in M_{m,n}(\mathbb{R})$, $B \in M_{n,p}(\mathbb{R})$, $C \in M_{p,q}(\mathbb{R})$, $D \in M_{q,m}(\mathbb{R})$ satisfy $A^T = BCD$, $B^T = CDA$, $C^T = DAB$ and $D^T = ABC$. Prove that $(ABCD)^2 = ABCD$.

4. (IMC) Let $n > 1$ be odd and let $A = (a_{ij})$ be the $n \times n$ matrix with $a_{ij} = 2$ for $i = j$, $a_{ij} = 1$ for $i - j \equiv \pm 2 \pmod{n}$ and $a_{ij} = 0$ otherwise. Find $\det A$.
5. (CIIM 2011) For $n \geq 3$ let $(b_0, b_1, \dots, b_{n-1}) = (1, 1, 1, 0, \dots, 0)$ and let $C_n = (c_{ij})$ be the $n \times n$ matrix with $c_{ij} = b_{(j-i) \bmod n}$. Show that $\det C_n = 3$ if $3 \nmid n$ and $\det C_n = 0$ otherwise.